

▲ M001 Week 5 — English Worksheet

Topic: First Pyramid · **Course:** Pyramid Problems · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
set size to 9
repeat size times:
  repeat size times:
    place block down
    move forward
  [turn and walk back to start of next row]
```

This builds the bottom layer of the pyramid. How long is each side?


```
set size to 9
while size > 0:
  [build one size × size square layer]
  move up by 1
  set size to size - 2
```

Why does the pyramid shrink by 2 each layer instead of 1?


```
set size to 9
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 2
```

How many layers does the pyramid have? List the sizes from bottom to top.

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to build a pyramid where each layer is **2 blocks smaller** than the one below.

```
set size to 9
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 1
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Each new layer must sit **on top** of the one below.

```
set size to 9
while size > 0:
    [build one size × size square layer]
    set size to size - 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent is supposed to **stop** building once the layer is gone (size 0 or smaller).

```
set size to 9
while size > -10:
    [build one size × size square layer]
    move up by 1
    set size to size - 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Build a pyramid **starting at size 7** (so the layers go $7 \rightarrow 5 \rightarrow 3 \rightarrow 1$). Then try once more with a **different material** — sandstone, gold, or your choice. When you finish, come back here.

Send a photo or video of your pyramid, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

The bottom layer is ... blocks long.

Each new layer gets smaller by ... because ...

The pyramid has ... layers in total.

The hardest part was ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you walk the camera around your pyramid. Talk through it like you are teaching someone who has never seen it. Try to use these words: **pyramid, layer, size, shrink, stack**.

1. Walk around the pyramid and show all 4 sides.
2. Read your **while** loop out loud and say what each part does.
3. Show one bug you hit and how you fixed it.
4. Say how many layers your pyramid has and how the size changes each time.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.